# CSS Day 3 — Fundamentals

Detailed theory notes for internship learning and revision.

## 1. Introduction to CSS

CSS (Cascading Style Sheets) controls the appearance and layout of HTML. HTML provides structure, CSS provides styling, and JavaScript provides behavior. CSS controls colors, fonts, backgrounds, borders, dimensions, spacing, and layout.

## 2. CSS Syntax

A CSS rule has a selector followed by declarations inside curly braces. A declaration consists of a property and a value.

## 3. Inline CSS

CSS written directly in an HTML element using the style attribute. Useful for quick, element-specific styling but difficult to maintain when overused.

## 4. Internal CSS

CSS written inside a style element, usually in the HTML head. It can style multiple elements on one page.

## 5. External CSS

CSS written in a separate .css file and linked to HTML. It keeps HTML cleaner and allows styles to be reused across pages. It is generally preferred for larger projects.

## 6. CSS Selectors

Element selectors target tags. Class selectors start with a dot and can be reused. ID selectors start with # and are intended to identify a particular element; IDs should generally be unique. The universal selector * targets all elements.

## 7. Colors

CSS supports color names, hexadecimal values, RGB, and RGBA. Hex colors commonly use #RRGGBB. RGB represents red, green, and blue; RGBA additionally includes alpha/transparency.

## 8. Backgrounds

background-color changes the background color. background-image can place an image behind content. Other background properties control size, position, and repetition.

## 9. Borders

Borders create a visible boundary around an element. A border can have width, style, and color. Common styles include solid, dashed, dotted, and double.

## 10. Fonts and Text Styling

font-family controls the typeface; font-size controls text size; font-weight controls thickness; font-style controls italic/normal appearance; color controls text color; text-align controls alignment; text-decoration controls decorations; text-transform changes capitalization.

## 11. Width and Height

width controls horizontal size and height controls vertical size. Common units include px, %, em, rem, vw, and vh.

## 12. Margin

Margin is the space outside an element's border. It can be set for all sides or individually with margin-top, margin-right, margin-bottom, and margin-left.

## 13. Padding

Padding is the space inside the border, between content and border. Individual sides can also be controlled.

## 14. CSS Box Model

The box model consists of content, padding, border, and margin. Padding is inside the border; margin is outside the border.

## 15. Applying CSS to Your Employee Form

Practice styling the page background and font, headings, form width and border, input dimensions and padding, and buttons. Focus on understanding what each property changes.

## 16. Quick Revision

CSS = styles HTML. Inline/Internal/External = ways to apply CSS. Selector = identifies what to style. Class = reusable selector. ID = identifier intended to be unique. color = text color. background-color = background color. border = boundary. font-family/font-size/font-weight = font styling. width/height = dimensions. margin = outside spacing. padding = inside spacing.